

Sai kumar k

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DATA SCIENTIST/GEN AI SPECIALIST

PROFESSIONAL SUMMARY:

- 6+ years of IT industry experience as a Data Scientist/Machine Learning Engg specialized in implementing advanced Machine Learning, Deep Learning and Generative AI (LLM) upon data from diverse domains.
- Expertise in understanding Business problems, proposing Analytics & Data science solutions, data gathering and understanding, modeling, validating the results, presenting the insights to stakeholders and deploying the solution.
- Work independently with R&D partners to understand requirements.
- Experiences in machine learning, data mining, structured and unstructured data analysis, and including feature extraction, algorithm development, text mining, computer simulation, data modeling, databases design, model evaluation.
- Experience with Statistical Analysis, Data Mining and Machine Learning Skills using Python and SQL.
- Strong practical understanding of statistical modeling and supervised/unsupervised machine learning techniques with keen interests in applying these techniques to the predictive analytics world.
- Good familiarity in the entire Data Science project life cycle, including Data Acquisition, Data Cleansing, Data Manipulation, Feature Engineering, Modeling, Evaluation, Optimization.
- Experience in problem solving, algorithmic development, data science, machine learning, statistical inference, predictive analytics.
- Hands on experience on clustering algorithms like K-means, NMF.
- Expertise in Model Development, Data Mining, Predictive Modeling, Descriptive modeling, Data Visualization, Data Clearing and Management, and Database Management.
- Experience using machine learning models such as random forest, KNN, SVM, logistic regressions and used packages such as NumPy, Scikit learn, pandas, etc., python respectively.
- Professional working experience in Machine Learning algorithms such as Linear Regression, Logistic Regression, Random Forests, Decision Trees, K-Means Clustering and Association Rules.
- Highly motivated team player with excellent Interpersonal and Customer Relational Skills, Proven Communication, Organizational, Analytical, Presentation Skills, and Leadership Qualities.
- In-depth experience in Python, PySpark, SQL, MongoDB, Scikit Learn, Amazon AWS, REST APIs, LINUX, GIT, Knime. Professional working experience in Machine Learning algorithms such as Linear Regression, Logistic Regression, Random Forests, Decision Trees, K-Means Clustering and Association Rules.
- Experience with Amazon Web Services (AWS) in planning, designing, implementing and maintaining system applications in AWS Cloud in Windows and Linux Environments.

TECHNICAL SKILLS:

Development Tool	Python, Jupiter, Spyder IDE
Python Packages	Langchain, LLM, Numpy, SciPy, Pandas, scikit-learn, Matplotlib, seaborn, statsmodels.
Techniques	Machine learning, Regression, Clustering, Data mining, Text mining.
Machine Learning	Naïve Bayes, Decision trees, Regression models, Random Forests, Time- series, K-means
Operating Systems	Microsoft Windows, Linux (Ubuntu)

Programming & Scripting languages	Python, SQL
Database	SQL.
Algorithm's Skills	Machine Learning, Neural Networks, Deep Learning, NLP, Prediction, Pattern Identification, Data / Text mining, Regression, Logistic Regression, Clustering, Classification, Statistical modeling
Data Science/Data Analysis Tools &	Generalized Linear Models, Logistic Regressions, Boxplots, K-Means, Clustering, Tensorflow, Langchain, Open AI

PROFESSIONAL EXPERIENCE:

Organization: FutureSoft India Pvt Ltd
Role: Gen AI Specialist | AI Engineer

Aug 2025 - Present

Organization: NTT DATA
Role: ML Engineer | AI Engineer
Responsibilities:

Dec 2024 - Apr 2025

- Developed advanced Natural Language Processing (NLP) techniques to identify signal strengths based on adverse events, leveraging OpenAI's state-of-the-art language models.
- Designed and implemented a scalable system for dynamic data extraction using LLMs, including GPT-3.5, GPT-4, Claude V3.5 Sonnet, integrated with LangChain and AWS services (Textract, Bedrock).
- Entity Extraction with LLMs, Implemented logic for entity extraction using GPT-3.5, GPT-4, and Claude V3.5, optimized prompt resolution, and token handling within output limits.
- AI-Powered Prompt Management, Developed and managed AI-driven prompts, leveraging advanced LLMs for efficient processing of large datasets, including text and bounding box extractions.
- Integrated Distributed Tracing, Embedded distributed tracing mechanisms for performance monitoring and debugging throughout the data extraction workflow.
- Seamless Integration with FastAPI: Ensured extracted data and entity information was processed and formatted for API consumption, enhancing the end-to-end workflow.
- Created intuitive and user-friendly applications and dashboards using Streamlit, enabling users to interact effectively with the NLP models and view results seamlessly.
- Developed and maintained REST APIs using FastAPI, ensuring efficient communication between the front-end applications and the machine learning models.
- Utilized Language Models to create effective and accurate prompts for identifying narrative cases in pharmacovigilance.
- Served as a domain expert in Pharmacovigilance, meticulously identifying and documenting named entities such as Positive Dechallenge, Negative Dechallenge, Positive Rechallenge, and Negative Rechallenge.
- Designed and implemented machine learning pipelines utilizing AWS Sagemaker, enhancing model integration and performance.
- Developed robust models for signal detection, employing Python Outlier Detection (PyOD) and advanced clustering algorithms.
- Used text simplification techniques (stopword removal, punctuation removal, lemmatization) for making a text easier to read and understand, while preserving its main ideas and approximate meaning.
- Performed lexical normalization for translating/transforming a non-standard corpus to a standard text

form.

- Performed Topic Modeling for identifying abstract Topics underlying in the collection of J&J's covid vaccine customer complaints data using various algorithms including LDA, NMF, KMeans.
- Developed Intent Detection tool using traditional NLP methods for capturing the semantics of chatbot transcripts of employees in the payroll division of J&J and assigning them to the correct label.
- Extracted top keywords for identifying the most relevant terms to describe the subject of the obtained intents from the model.
- Created Text Summarization pipeline for shortened version of several documents belonging to the identified intents that preserves most of their meaning using cosine similarity and Transformers T5 Base models (Hugging Face).
- Utilized AWS Machine Translator for translating chatbot responses of employees from 10 different languages to English. Used text simplification techniques (stopword removal, punctuation removal, lemmatization) for making a text easier to read and understand, while preserving its main ideas and approximate meaning.
- Tagged entities in text with their belonging to their respective topics.
- Assisted Entity Linking to the content strategy team for recognizing and disambiguating named entities, designing phrases, intents, and solutions to the knowledge base of J&J chatbot system.

Environment: Python, LLM, Open-AI, Tfidf, Hugging-face Transformers, spaCy, NLTK, Gensim, coreNLP, TextBlob, Pytorch, NLP, TensorFlow, BLSTM, scikit-learn, Pandas, Numpy, Regex, aws.translate, time series models, ARIMA, SARIMA.

Organization: Swiftsoft Infotech Services Pvt Ltd
Role: Software Engineer

Jan 2022 - Dec 2024

Project: Topic Modelling Description: This project is related to a big Pharmaceutical company, As per the business requirement, we have to work on "JANSSEN COVID-19 VACCINE VIAL" product, where we have to find out customer's feedback/quires after getting COVID vaccine and categories them so that ChatBot can understand and respond back to the candidates. Here we have client's data around 2 million in csv format in which we have to put our logic and find out the structured data by using KMeans algorithm.

Responsibilities:

- Utilized AWS Machine Translator for translating chatbot responses of employees from 10 different languages to English.
- Used text simplification techniques (stopword removal, punctuation removal, lemmatization) for making a text easier to read and understand, while preserving its main ideas and approximate meaning.
- Performed lexical normalization for translating/transforming a non-standard corpus to a standard text form.
- Performed Topic Modeling for identifying abstract Topics underlying in the collection of J&J's covid vaccine customer complaints data using various algorithms including LDA, NMF, KMeans.
- Developed Intent Detection tool using traditional NLP methods for capturing the semantics of chatbot transcripts of employees in the payroll division of J&J and assigning them to the correct label
- Extracted top keywords for identifying the most relevant terms to describe subject of the obtained intents from model.
- Created Text Summarization pipeline for shortened version of several documents belonging to the identified intents that preserves most of their meaning using cosine similarity and Transformers T5 Base models (Hugging Face)
- Tagged entities in text with their corresponding type, typically in BIO notation using Named Entity Recognition techniques including StanfordNLP, count vectorizer packages.
- Built Phrases Generation pipeline using n-grams, fuzzywuzzy, sentence similarity models for identifying the top common phrases belonging to their respective topics.
- Assisted Entity Linking to the content strategy team for recognizing and disambiguating named entities, designing phrases, intents, and solutions to the knowledge base of J&J chatbot system.
- Created an API by utilizing Flask and imparted the plan to application group and enable them to characterize the models

Environment: Python, Tfidf, stancorenlp, Hugging-face Transformers, spaCy, NLTK, Gensim, coreNLP, TextBlob, Pytorch NLP, TensorFlow, BLSTM, scikit-learn, Pandas, Numpy, Regex.

Project:

Implementation of Spider Crawler for finding Job information in Organization's career portal with specialized Browser Based Automation System functionalities which works of All the kind of features on Extracting the Job Profile information. Performing Analytic on Job Portal data by using WMD, NLTK Toolkit & NLP Process on using Python Technology.

Responsibilities:

- Collecting the Organization information form Job Search Engines
- Used python for web scrapping (Beautiful Soup, Selenium, scrapy, Puppeteer)
- Building Structured Data Finder
- Extracts all the Job Details from various company websites
- Classifying similar Jobs into various buckets based on Job Description, Skills, Professional • Expectations, Educational background, etc.
- Extracted Job Details stored as time series in Mongo dB and other supporting data stores
- Used NLTK tool for Sentimental Analysis
- Building User Interface with Django Framework
- Finding the Similarities between description using metric as "Word Movers Distance" under genism module.
- Creating AWS Lambda Function & Deploying Python script in AWS.
- Creating REST API with AWS API Gateway's

Organization: AVSAR HR Services

Nov 2019 - Dec 2021

Role: Business Development Executive

Project: Sales Call Quality Monitoring and Compliance Enhancement in Pharma Sales

Responsibilities:

- Designed and maintained MIS reports covering sales, purchases, returns, and outstanding payments.
- Created a standardized template for sales call tracking, including order details, customer feedback, and delivery issues.
- Coordinated daily with customers to take orders, confirm stock availability, and assist with returns or damaged items.
- Improved order accuracy and return processing time by 50% through structured follow-up and QA monitoring.
- Collaborated with the accounts team for reconciliation of F & F settlements and timely processing of customer refunds.
- Trained new team members on call protocols, Excel tracking, and QA checklists.

Organization: V-Dream Technology Communication PVT LTD

March 2018 - Oct 2019

Role: VBA

Responsibilities:

- Greet and assist walk-in customers in selecting suitable mobile phones or accessories.
- Understand customer needs and recommend appropriate products and offers.
- Upsell and cross-sell mobile devices, data plans, and accessories.
- Explain product features, specifications, and pricing clearly.
- Stay updated on the latest mobile brands, models, features, and market trends.